Instinct Lifecycle & Semantic Search — Design Sketch

Status: DRAFT for review — nothing wired, nothing committed **Date:** 2026-07-10 **Author:** Alaric (for marlandoj)

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1. Why we're here

The daily `[MEM] Decay Memory` agent (Memory Pipeline B) prunes and synthesizes the episodic/semantic memory store. The open question was whether instinct knowledge should ride along. Tracing the read path answered a prior question first: **instincts are not unwired.** They fire on every prompt via `memory-gate-hook.sh`, which calls `observer.ts brief` and injects an `<instincts>` block. The `[Session Briefing — Instincts]` you see at the top of prompts *is* the system working.

So this is not "turn instincts on." It's "give the corpus that already exists a maintenance lifecycle and better retrieval." Two gaps, two phases.

The two gaps

Gap	Today	Consequence
No lifecycle	`prune.ts` is manual-only; cap 200 never hit (123 live); confidence is **monotonic** (`Math.max` on reinforce — only ratchets up, never down)	Stale or contradicted instincts persist at full strength. Confidence is a high-water mark, not a live reliability signal.
Crude retrieval	`selectForBriefing` floats instincts whose `domain` substring literally appears in the prompt, then slices top-N by confidence	Keyword-only. A prompt about "the conveyor" won't surface a `factory`-domain instinct unless the word "factory" appears. No semantics.

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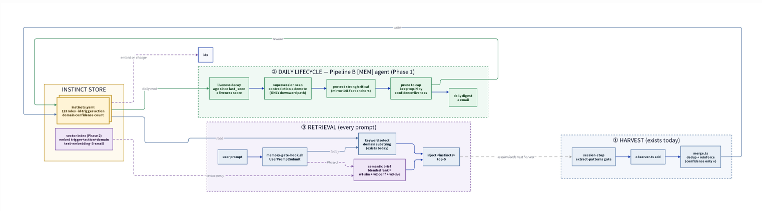
2. The shape of the answer

You were partial to prune/decay; you also saw value in semantic search; and you asked whether it has to be a tradeoff. **It doesn't.** They operate on different axes and compound rather than compete:

- **Lifecycle** decides *what deserves to exist and at what strength* (write/maintain side).
- **Semantic index** decides *what surfaces for a given prompt* (read side).

A healthy lifecycle makes semantic results trustworthy (you're searching a pruned, liveness-weighted corpus, not a junk drawer). Semantic retrieval makes a well-maintained corpus actually reachable. Do lifecycle **first** — it's the foundation, and it validates your own instinct to start with prune/decay.



Instinct Lifecycle Workflow

Grey = exists today · Green = Phase 1 (lifecycle) · Purple = Phase 2 (semantic).

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3. Phase 1 — Confidence Lifecycle (foundation)

Runs inside Pipeline B, once daily, right alongside memory decay. Advisory-first (report before it mutates), same discipline as every prior factory phase.

3.1 Liveness decay

A **liveness score** separate from confidence, derived from `last_seen` age. Recent, repeatedly-reinforced instincts stay live; ones untouched for a long window fade.

- Confidence stays a **quality** signal; liveness is a **recency** signal. We never conflate them.
- Decay is gentle and continuous, not a cliff.

3.2 Supersession demotion — the *only* downward confidence path

Today nothing can lower confidence. That's wrong for a rule that's been *contradicted* (not just gone quiet). We mirror the memory system's supersedes-aware machinery (P1-4): when a newer instinct contradicts an older one on the same trigger, the older one is **demoted**, and the newer one carries a pointer to what it replaced.

- Two kinds of "change over time," handled separately:

    - **Staleness** (instinct went quiet) → liveness decay.
    - **Contradiction** (instinct is now wrong) → supersession demotion.
- This is the piece that makes “instincts change over time” safe.

3.3 Protection

Strong, high-confidence, or explicitly-critical instincts are **protected** from eviction — the same pattern as fact-decay’s 141 protected articulation points. Your “stronger instincts always surface” requirement is a first-class rule, not a side effect.

3.4 Prune

After liveness + supersession + protection, prune to cap by a blended keep-score (confidence · liveness), protecting anchors. Turns the manual `prune.ts` into a scheduled, principled step.

3.5 Daily digest

A short report (what decayed, what was demoted/superseded, what was pruned, what’s protected) emailed with the Pipeline B run so you see the corpus breathing.

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4. Phase 2 — Semantic Index (retrieval upgrade)

Only after Phase 1 is trusted.

4.1 Embed the corpus

Embed each instinct’s `trigger + action + domain` with `text-embedding-3-small` (same model already used in the memory runtime) into a searchable vector index, kept in sync on write/lifecycle changes.

4.2 Blended-rank brief

`observer.ts brief` gains a semantic path: rank by

```
score = w1·similarity + w2·confidence + w3·liveness
```

so retrieval blends *relevance to this prompt*, *quality*, and *recency*. The keyword path stays as a cheap fallback (and A/B baseline). The `<instincts>` injection contract is unchanged — only the selection improves.

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5. Sequencing & rollout

1. **Phase 1** behind flags, advisory-only, run inside Pipeline B for ~1 week; watch the daily digest; confirm

decay/supersession/protection behave before anything mutates for real.
2. **Phase 2** once the maintained corpus is trustworthy; ship keyword vs. semantic as an A/B so we can measure lift, not assume it.
3. Every step byte-identical when flags are off — the standing factory rule.

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6. Open decisions (yours to make)

- **Decay window / half-life** — how many quiet days before liveness meaningfully fades?
- **Protection threshold** — confidence floor (e.g. ≥0.90) and/or an explicit `critical` tag?
- **Cap** — keep 200, or right-size to the live corpus (123)?
- **Digest cadence** — every Pipeline B run, or only on material change?
- **Blend weights** — starting `w1/w2/w3` for the semantic score.

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7. What is NOT happening yet

No code wired, no schema changed, no Linear issue filed. This is a sketch for your review. On your go-ahead I'll formalize it into a Linear project (Phase 1 / Phase 2 issues) and begin Phase 1 advisory-only.